Rim-Guided Bowl Grasping

From a segmented point cloud to an executable grasp pose

INPUT

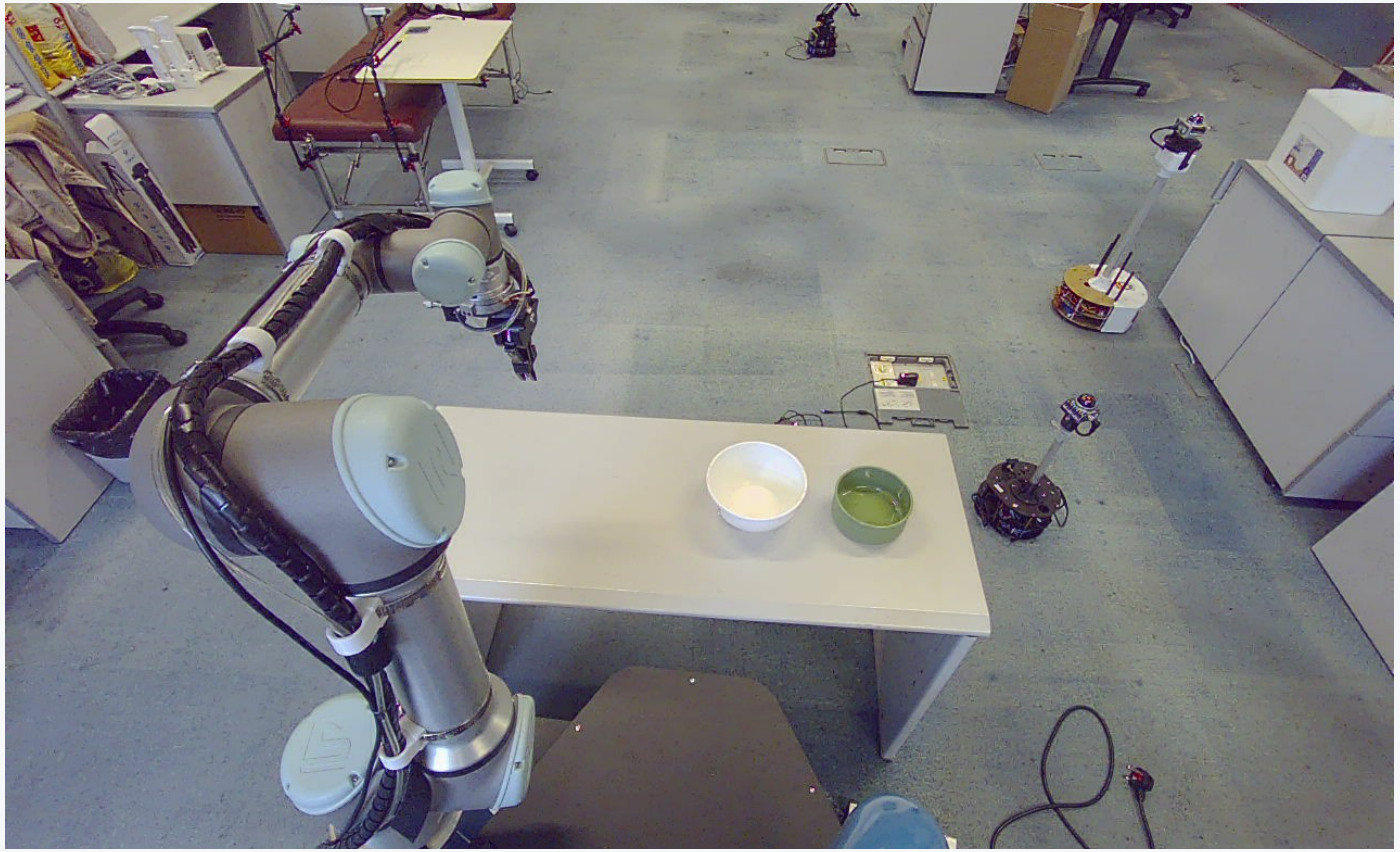
Instance-labelled RGB-D points
+ acquisition-time robot TF

CORE CONTRIBUTION

A lightweight, geometry-based symbolic model of rim contact

OUTPUT

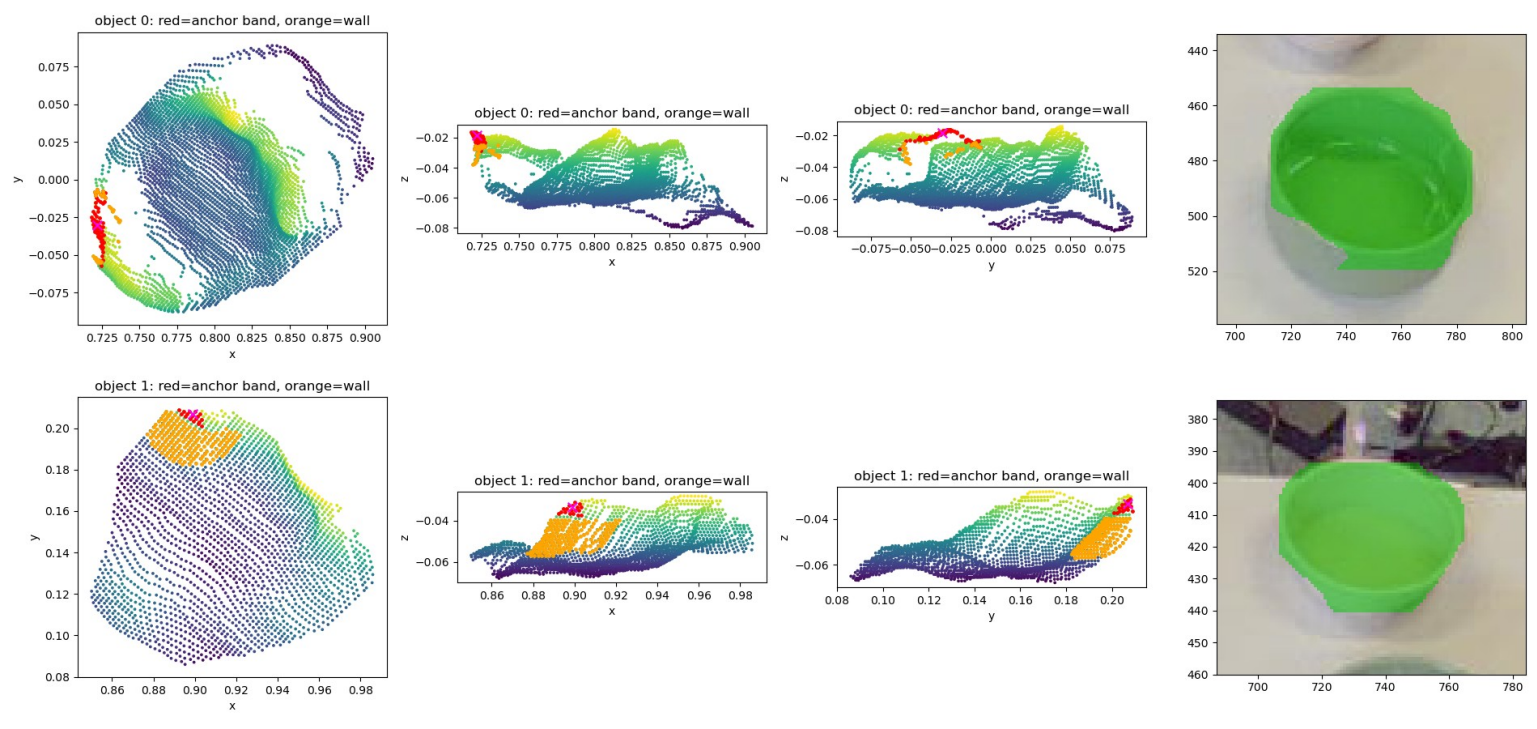
XYZ + orientation + diagnostics for downstream MoveIt planning



Sources: S1, S3, S4, S5 | Original lab image: I1 (27 Sep 2026)

A detected bowl is not yet a supported grasp

The original height-band method was simple, but local wall evidence could fail.



Recorded single-patch diagnostic: rim points in red; wall support in orange.

ORIGINAL METHOD

98th-percentile height

15 mm top band

Near-side median → XYZ

OBSERVED FAILURE

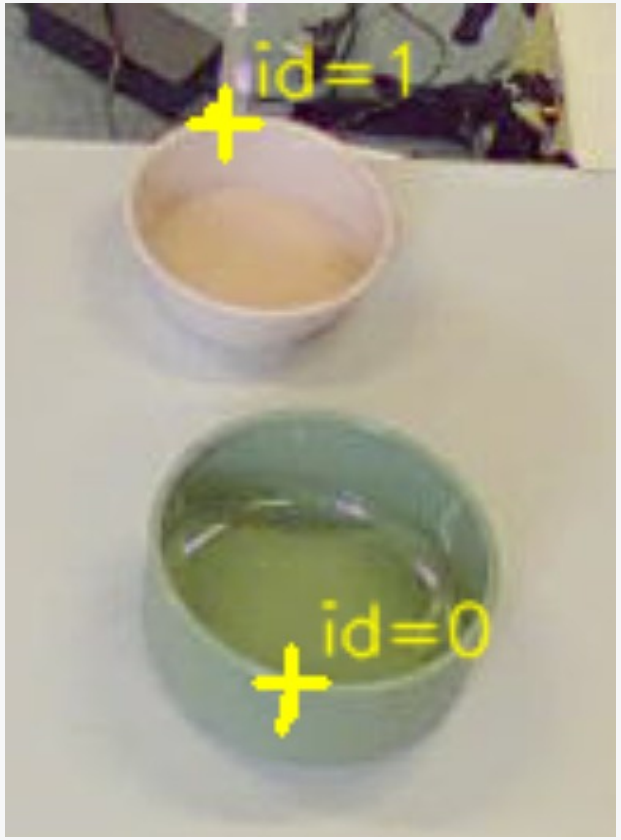
Two masks detected;
both wall fits rejected
in one recorded snapshot.

Segmentation alone cannot validate a rim contact.

Sources: S1, S2 | Original point-cloud diagnostic: I2 (23 Sep 2026)

Accept contacts only when the wall supports them

Same geometric rules for both bowls; no colour-specific model.



Projected contacts on a real capture

01 ADAPT THE RIM BAND

b = clip(0.20 × H, 6, 12) mm

H: robust height span (mm); upper-band anchor.

02 VERIFY THE LOCAL WALL

Fit at 12.5 / 18.75 / 25 mm radii.

Require agreement at two supported scales.

≥20 points • ≥6 mm height span • ≤2.5 mm fit residual

03 BUILD THE TOOL POSE

Tool Y: closing direction. Tool Z: downward.

Quaternion for control; RPY for inspection.

Current demo: tool tilt capped at 0°; measured wall tilt is logged separately.

Sources: S2, S5, S6, S10 | Original projected-contact image: I3

From a rim point to a complete pick-and-place task

Geometry stays in rim_locator; orchestration and motion remain separate.

rim_locator

Up to 8 poses per bowl; 4 mm separation.

Versioned groups + per-candidate diagnostics.

ROS interfaces

Preserve instance IDs, source time and TF.

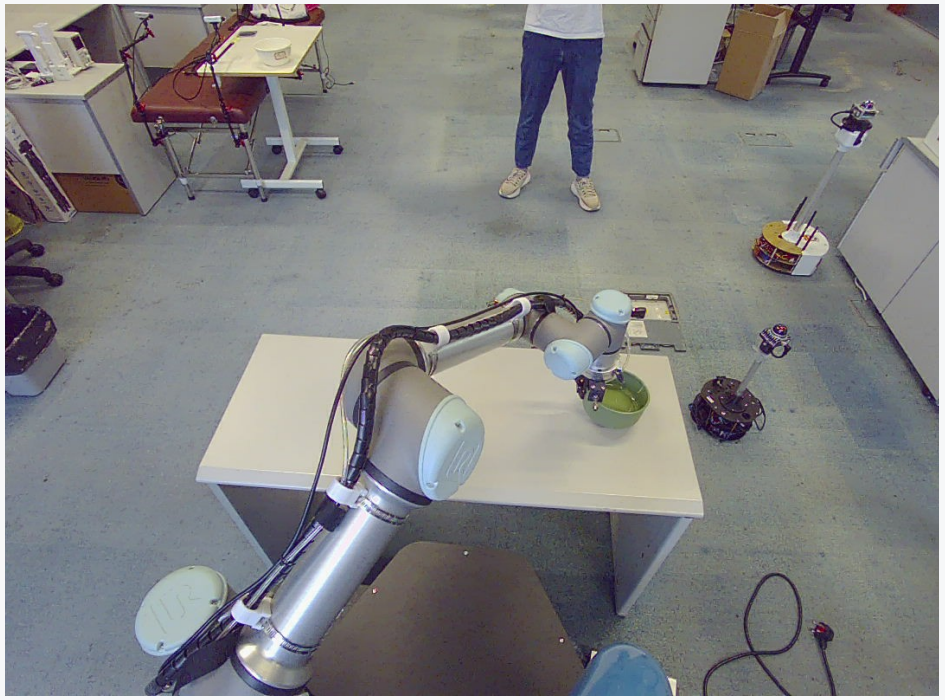
Keep PointStamped / PoseArray compatibility.

Execution & supervision

Try alternatives before any motion.

Re-observe after release and return; stop on execution failure.

10 cm approach/lift | 20 cm placement drop | 1 s dwell



Original green-bowl grasp image • 27 Sep

Fixed placement pose;

nearest bowl first in each new observation.

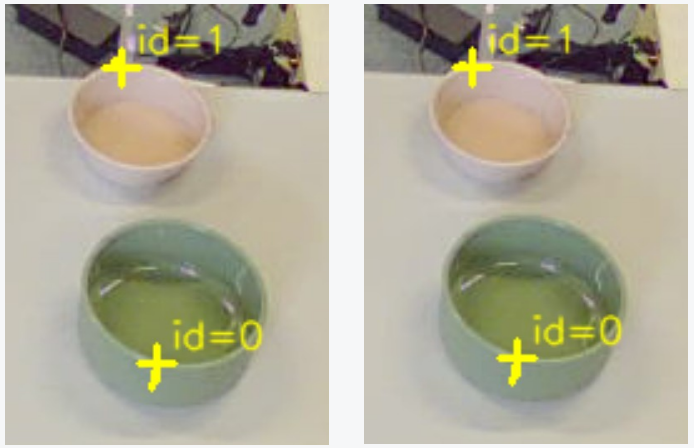
Sources: S3, S4, S6, S12 | Original image: I4 | Candidate fallback is pre-motion only.

Real captures support the geometric design

Three recorded two-bowl observations; replay results are not a grasp success rate.

Capture	Accepted	Green tilt	Light tilt	Residual*
01	**2 / 2**	11.73°	31.89°	0.55–0.61 mm
02	**2 / 2**	13.82°	36.80°	0.60–0.62 mm
03	**2 / 2**	8.82°	37.37°	0.46–0.74 mm

* Local plane-fit residual, not independently measured localization accuracy.



Original contact projections
Capture 01 / Capture 03

503

Regression tests passed
27 Sep record

15

MoveIt integration checks
25 Sep; simulated control

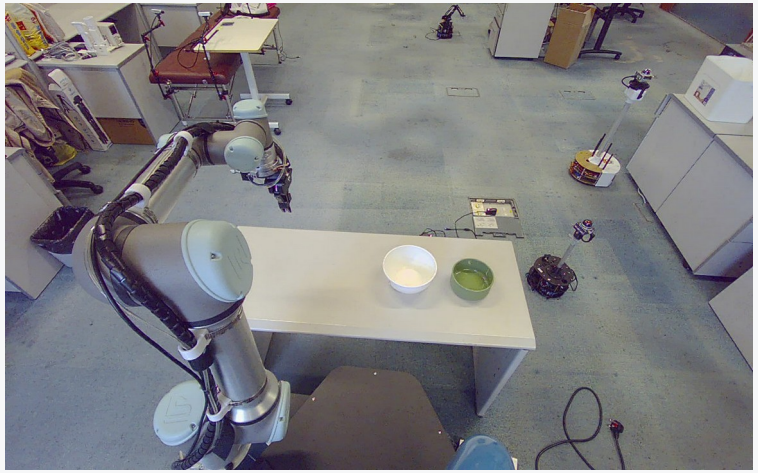
26

Historical replay checks
25 Sep record

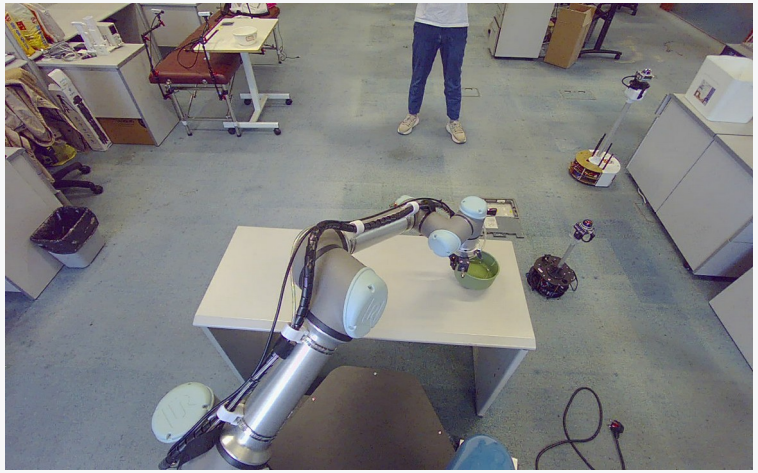
Sources: F1–F3, S2, S7, S8, S11 | Original images: I3, I7 | All test counts are dated records.

One complete cycle; reliability remains to be measured

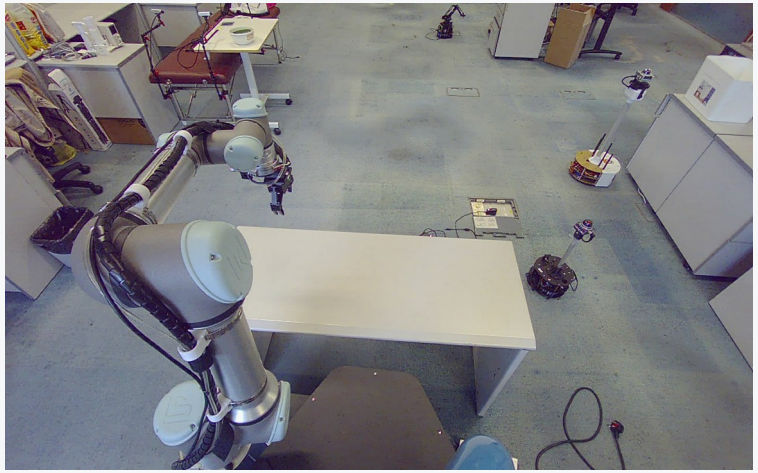
Documented case study: 27 September 2026



Initial source scene



Green-bowl grasp



Final source scene

AUDITED OUTCOME

Green: 1 complete cycle + fresh empty-ROI observation.
White: lifted/placed; retreat cancelled → incomplete.

NEXT VALIDATION

Controlled repeats across bowls and poses.
Measure physical success; add grasp/release feedback.

Sources: S4, S8, S9 | Original images: I1, I4, I5 | No overall physical success rate claimed.